

MONDY, Russian Federation

WMO: 308020

Lat: 51.683N

Lon: 100.983E

Elev: 1304

StdP: 86.61

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-35.1	-32.5	-37.0	0.1	-34.8	-34.5	0.2	-31.9	8.5	-13.7	7.6	-12.0	0.2	290	0.696

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	25.4	14.7	23.5	14.2	21.7	13.7	16.7	22.1	15.6	20.7	14.7	19.7	2.9	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.7	12.3	19.2	13.7	11.4	17.7	12.7	10.7	16.7	52.1	22.1	48.6	20.5	45.7	19.7	21.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.8	6.8	5.9	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-1.5	-18.7	-16.9	-8.5	-0.3	6.0	12.5	15.3	12.6	6.1	-1.0	-10.0	-15.9
	DBStd	12.68	5.51	6.00	5.99	4.94	4.04	3.58	2.55	2.88	3.51	4.75	5.99	5.98
	HDD10.0	4557	889	753	573	310	136	17	1	11	123	341	599	804
	HDD18.3	7234	1148	987	832	559	383	176	97	177	366	599	849	1062
	CDD10.0	369	0	0	0	1	11	92	166	93	8	0	0	0
	CDD18.3	5	0	0	0	0	0	1	4	0	0	0	0	0
	CDH23.3	176	0	0	0	0	10	57	91	17	1	0	0	0
	CDH26.7	21	0	0	0	0	1	8	12	0	0	0	0	0
Wind	WSAvg	2.1	1.8	1.8	2.4	2.7	2.7	2.0	1.7	1.7	2.1	2.0	2.1	2.0
Precipitation	PrecAvg	334	2	2	4	11	23	63	102	76	34	8	4	3
	PrecMax	471	5	7	9	23	48	134	192	117	70	26	12	12
	PrecMin	220	0	0	0	2	6	15	55	24	4	2	1	0
	PrecStd	71	1	2	2	6	10	25	30	26	19	5	2	2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-1.8	2.0	10.6	17.8	25.1	28.0	28.2	25.2	21.8	15.0	8.0	-0.5
		MCWB	-4.9	-3.0	2.6	7.7	11.4	15.2	17.3	14.2	11.2	5.9	2.3	-4.1
	2%	DB	-5.4	-1.4	7.7	13.9	20.9	25.1	25.7	23.0	18.5	12.1	4.5	-2.9
		MCWB	-8.0	-5.2	1.2	5.6	9.9	13.6	15.9	14.6	10.5	4.3	-0.2	-5.9
	5%	DB	-7.9	-4.1	5.0	11.4	18.2	22.7	23.8	21.2	16.0	9.6	1.8	-4.7
		MCWB	-9.9	-7.1	-0.7	3.8	8.5	13.4	15.4	13.9	8.9	3.5	-1.9	-7.3
	10%	DB	-10.3	-6.7	2.3	9.1	15.6	20.7	21.8	19.2	13.4	7.5	-0.7	-6.7
		MCWB	-12.0	-9.1	-2.4	2.5	7.1	12.6	14.7	13.2	7.7	2.4	-3.7	-8.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-4.7	-2.6	3.1	8.1	12.7	16.9	19.1	16.8	13.3	6.8	2.5	-3.9
		MCDB	-1.8	1.8	10.0	16.0	22.9	24.2	26.0	22.4	18.6	12.7	7.5	-1.0
	2%	WB	-7.9	-5.1	1.3	6.0	10.6	15.2	17.4	15.6	11.5	5.0	-0.1	-5.9
		MCDB	-5.6	-1.5	7.6	13.2	19.5	21.6	23.0	21.0	16.2	10.7	4.2	-3.2
	5%	WB	-9.9	-7.0	-0.5	4.1	9.1	14.2	16.4	14.7	10.1	3.8	-1.7	-7.2
		MCDB	-8.0	-4.2	4.7	10.3	16.8	20.2	21.1	19.3	14.5	9.1	1.7	-4.8
	10%	WB	-11.9	-9.0	-2.2	2.8	7.7	13.2	15.4	13.7	8.4	2.5	-3.5	-8.8
		MCDB	-10.3	-6.7	2.1	8.7	14.5	19.4	20.2	17.8	12.7	7.1	-0.9	-6.8
Mean Daily Temperature Range	5% DB	MDBR	13.3	16.3	15.9	12.7	12.6	11.9	11.1	11.0	12.2	13.2	12.7	11.9
		MCDBR	14.4	17.7	18.7	17.3	18.8	17.6	15.6	15.8	17.9	17.8	14.0	12.0
		MCWBR	12.3	14.3	12.8	10.2	9.6	8.0	7.0	7.5	9.5	10.7	9.9	9.8
	5% WB	MCDBR	14.3	17.6	17.4	15.8	16.0	14.1	12.6	12.5	14.6	16.7	13.8	11.9
		MCWBR	12.2	14.2	12.0	9.8	8.9	7.1	6.7	6.7	8.3	10.3	9.8	9.9
Clear-Sky Solar Irradiance	taub	0.232	0.258	0.301	0.347	0.379	0.373	0.369	0.343	0.303	0.281	0.253	0.232	
	taud	2.237	2.200	2.140	2.195	2.200	2.346	2.400	2.447	2.501	2.427	2.270	2.193	
	Ebn at Noon	801	871	890	887	872	880	878	886	892	842	758	724	
	Edh at Noon	65	91	119	129	135	118	110	100	84	73	63	57	
All-Sky Solar Radiation	RadAvg	1.33	2.32	3.77	5.14	5.92	5.69	5.30	4.68	3.68	2.43	1.39	0.99	
	RadStd	0.05	0.07	0.17	0.22	0.33	0.38	0.39	0.27	0.25	0.14	0.08	0.05	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-181	N/A	N/A

Nomenclature: See separate page